engPENK Amendment 4 — protease liability retracted, option B recommended

Supersedes: three statements in Amendments 1–3. **Basis:** five primary sources, cited inline. **Status of prior amendments:** everything not named in §0 stands.

§0 Retractions

R1 — the NEP site count was wrong (Amendments 1, 2, 3). I counted every bond followed by a hydrophobic residue as a neprilysin site, which made `YGGFEVVG` look like a 3-site peptide against Met-enkephalin's 1. Neprilysin's S1′ pocket is shaped by Phe563 and imparts a preference for **aromatic** P1′ residues; a valine at P1′ is too small and leaves an unfavourable gap on binding (Sexton et al. 2012, PLoS ONE 7:e32343). Val-P1′ bonds are therefore not equivalent sites.

Corrected count — **every species carries exactly one canonical NEP site, `Gly3-Phe4`, the same bond as native enkephalin**:

peptide	shipped count	canonical (aromatic P1′)	poor (Val P1′)
Met-enkephalin `YGGFM`	1	1	0
engPENK octapeptide `YGGFEVVG`	3	**1**	2
option A `YGGFEVV`-NH_2	2	**1**	1
option B `YGGFEVVG`-NH_2	3	**1**	2
engMERF `YGGFEVVGRF`	3	**1**	2

R2 — "amidation blocks ACE" was wrong (Amendment 1 table). ACE cleaves substrates whose C-terminus is amidated, including dipeptide release from substance P and CCK-8 and tripeptide release from GnRH (Bernstein et al. 2013, Pharmacol Rev 65:1). Amidation slows ACE; it does not confer immunity. Both option A and option B are ACE substrates.

R3 — "the cassette is less protease-stable than native enkephalin" does not survive R1+R2. With one canonical NEP site each and both species ACE-accessible, there is no evidence base for a stability penalty. I am not replacing it with a claim of greater stability — the Val-P1′ bonds are disfavoured, not proven inert, and no kinetics on these sequences exist. The honest position: **NEP/ACE liability is comparable to native, and measuring it is a wet-lab task, not a sequence-inspection task.**

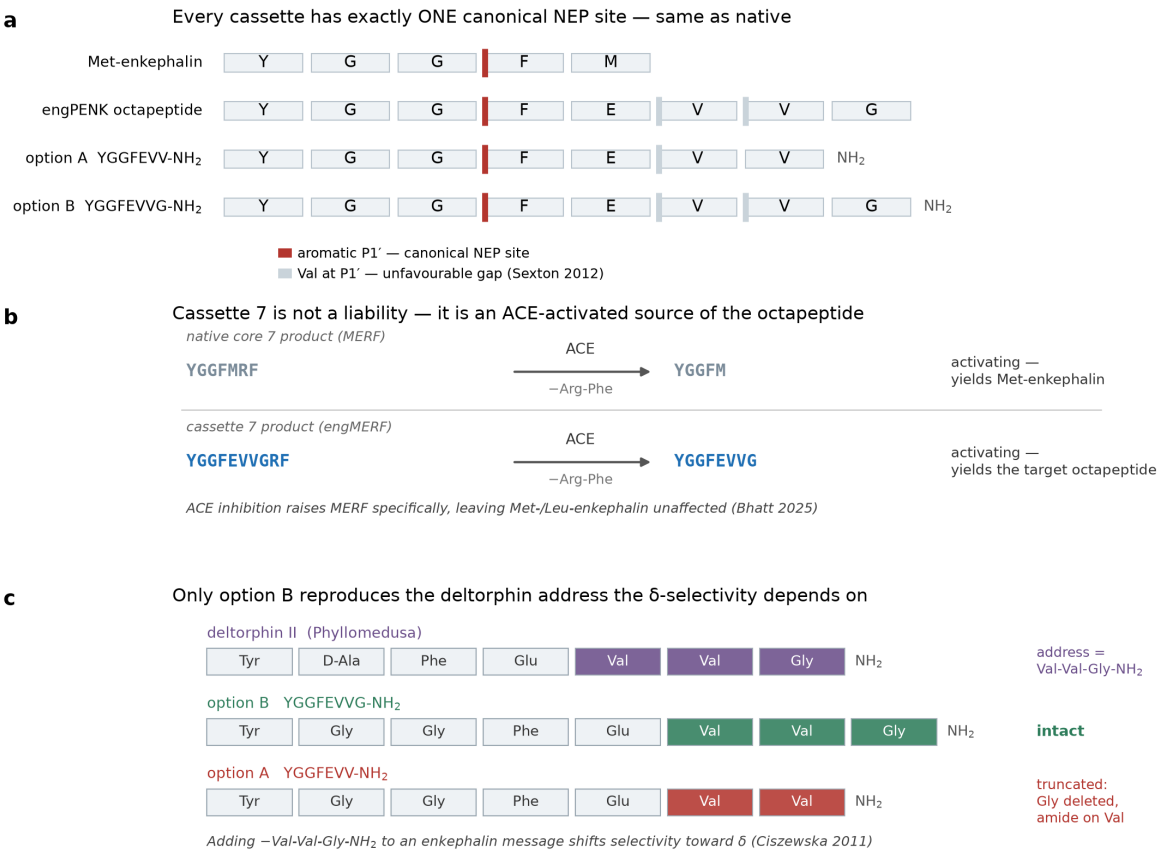


Figure. (a) Tiered NEP subsite scoring — one canonical aromatic-P1′ site per species. (b) ACE dipeptide removal converts the cassette 7 product into the target octapeptide, mirroring the native MERF→Met-enkephalin conversion. (c) Option B preserves the deltorphin -Val-Val-Gly-NH_2 address; option A truncates it.

§1 Cassette 7 is an asset, not a defect

ACE is a dipeptidyl carboxypeptidase. On native MERF (`YGGFMRF` , core 7's product) it removes Arg-Phe to give Met-enkephalin — an activating conversion. Applied to the cassette 7 analogue:

substrate	ACE removes	product	[M+H]⁺
`YGGFMRF` (native)	Arg-Phe	`YGGFM`	574.2330
`YGGFEVVGRF` (cassette 7)	Arg-Phe	`YGGFEVVG`	**827.3934**

The cassette 7 product is converted by ACE into the target octapeptide, by the identical chemistry that activates the native peptide. Independently, ACE inhibition raises extracellular MERF specifically while leaving Met- and Leu-enkephalin unaffected (Bhatt et al. 2025, Commun Biol), which means the MERF→enkephalin route is ACE-dependent in vivo, not a test-tube artefact. Amendment 2 treated cassettes 4 and 7 as an open pharmacology question; for cassette 7 the answer now leans positive.

§2 Option B is the recommended cassette

Deltorphin II is `Tyr-D-Ala-Phe-Glu-Val-Val-Gly`-NH_2. Its δ-selectivity is attributed to a C-terminal **address** sequence, named explicitly as `-Val-Val-Gly`-NH_2; appending that address to an enkephalin message shifts receptor selectivity toward δ (Ciszewska et al. 2011, Acta Biochim Pol 58:225 — a construct architecturally identical to this project's: enkephalin message elongated by a deltorphin C-terminal fragment).

- **option B** `YGGFEVVG`-NH_2 — carries `-Val-Val-Gly`-NH_2 **intact**
- **option A** `YGGFEVV`-NH_2 — deletes the address glycine and relocates the amide onto Val

Option A does not merely shorten the peptide; it destroys the motif the δ-selectivity rationale rests on. **Recommendation: option B.**

Two cautions. First, Ciszewska et al. note the selectivity shift generally arises from losing μ affinity rather than gaining δ affinity — expect a selectivity ratio change, not a potency gain. Second, deltorphin's message uses D-

Ala²; the cassette necessarily uses the genetically encodable `Tyr-Gly-Gly-Phe`, so the message domain is enkephalin's, not deltorphin's. Option B is an enkephalin-message/deltorphin-address hybrid, which is exactly the class Ciszewska characterised — but its affinities are unmeasured.

§3 What this changes on the bench

1. **Order option B** (`YGGFEVVG`-NH_2, i.e. cassette encodes `...YGGFEVVGG-KR...`). The Gly is the PAM substrate and the address residue — one glycine serving both roles.
2. **Run the Ozawa 3-step cascade on** `YGGFEVVGGKR` (Amendment 3 §2), not `YGGFEVVGKR`. The −58.01 Da amidation readout is unchanged.
3. **Add an ACE arm to the MS panel.** Expect `YGGFEVVG` (827.3934) from the cassette 7 product and `YGGFEV` (671.3035) from over-digestion of the octapeptide itself.
4. **Do not report a stability advantage** for the cassette in any grant text or figure legend until NEP/ACE kinetics on `YGGFEVVG` are measured.

§4 Checklist

- [ ] Peptide order switched to option B (`YGGFEVVG`-NH_2) and the +Gly cassette sequence
- [ ] Amendment 1 liability table reissued with the corrected NEP counts (R1) and ACE column (R2)
- [ ] Synthetic `YGGFEVVGGKR` ordered for the in-vitro cascade
- [ ] ACE (± captopril) arm added to the degradation protocol
- [ ] Any slide or draft claiming improved protease stability corrected

References

1. Sexton T. et al. (2012) PLoS ONE 7:e32343 — NEP S1′ pocket, Phe563, P1′ preference. doi:10.1371/journal.pone.0032343
2. Bernstein K.E. et al. (2013) Pharmacol Rev 65:1 — ACE dipeptidyl carboxypeptidase; amidated substrates. doi:10.1124/pr.112.006809
3. Bhatt et al. (2025) Commun Biol — ACE inhibition raises MERF selectively. doi:10.1038/s42003-025-08790-6
4. Ciszewska M. et al. (2011) Acta Biochim Pol 58:225 — enkephalin/deltorphin hybrids; `-Val-Val-Gly`-NH_2 address. doi:10.18388/abp.2011_2269
5. Ozawa A. et al. (2007) Anal Biochem — in-vitro PC2/CPE/PAM cascade. (Amendment 3 ref.)